NOTE: The document identifier and heading have been changed on this page to reflect that this is a performance specification. There are no other changes to this document. The document identifier on subsequent pages has not been changed, but will be changed the next time this document is revised.

MIL-PRF-39022/7C
19 July 1984
SUPERSEDING
MIL-C-39022/7B
16 June 1972

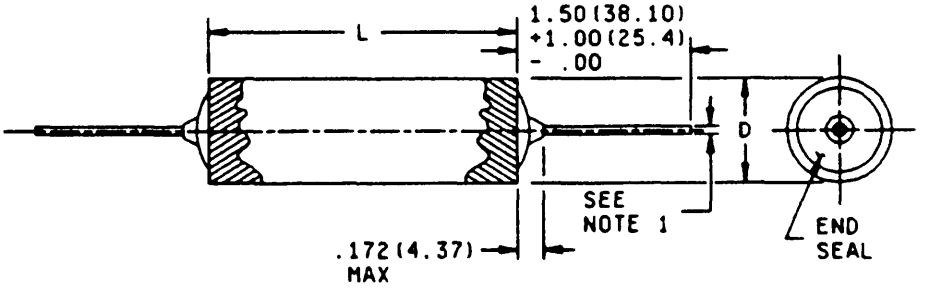
PERFORMANCE SPECIFICATION SHEET

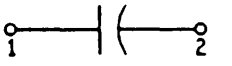
CAPACITORS, FIXED, PLASTIC DIELECTRIC (METALLIZED PLASTIC FILM)
ALTERNATING CURRENT, (RMS)
(HERMETICALLY SEALED IN METAL CASES)
ESTABLISHED RELIABILITY
STYLE CHR49 (UNINSULATED)

FAILURE RATE LEVEL "L" IS INACTIVE FOR DESIGN AFTER 20 JULY 1983

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the capacitors described herein shall consist of this specification sheet and the issue of MIL-C-39022.





CIRCUIT DIAGRAM

NOT TO BE USED WITH INSULATING SLEEVES.

NOTES:
1. Number 20 AWG wire (.032 inch ±.002 (0.81 ±0.05 mm)) for case diameter of .312 (7.92 mm)
   Number 18 AWG wire (.040 inch ±.002 (1.02 ±0.05 mm)) for case diameter of .400 (10.16 mm) and over.
2. See table I for additional dimensions.
3. Dimensions are in inches.
4. Metric equivalents are given for general information only.
5. Metric equivalents are in parentheses.
8. Lead length may be a minimum of 1 inch (25.4 mm) long for use in tape and reel packaging when specified in the ordering data.

FSC 5910

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REQUIREMENTS:

Dimensions and configuration: See figure 1 and table I.

Case material: Nonmagnetic.

Capacitance value: See table I.

Capacitance tolerance: 10 percent.

Rated voltage: RMS sine wave ac at 60 through 400 Hz (see table I).

Dielectric material: Normally polycarbonate.

Operating temperature range: -55°C to +85°C.

Failure rate level: M(1.0%), P(0.1%), R(0.01%), and S(0.001%).

TABLE I. Electrical characteristics, dimensions, and dash numbers.

AC rated voltage volts (rms)	Capacitance value µF	Dimensions, 1/ nominal (inches)		Dash number Failure rate level for (%/1,000 hr)			
		D ±0.062	L +.031 -.051	M (1.0)	P (0.1)	R (0.01)	S (0.001)
400	.01	0.312	1.125	1171	1191	1211	1231
400	.012	0.400	1.125	1270	1290	1310	1330
400	.015	0.400	1.125	1172	1192	1212	1232
400	.018	0.400	1.125	1271	1291	1311	1331
400	.022	0.400	1.125	1173	1193	1213	1233
400	.027	0.400	1.375	1272	1292	1312	1332
400	.033	0.400	1.375	1174	1194	1214	1234
400	.039	0.562	1.125	1273	1293	1313	1333
400	.047	0.562	1.125	1175	1195	1215	1235
400	.056	0.562	1.355	1274	1294	1314	1334
400	.068	0.562	1.355	1176	1195	1215	1236
400	.082	0.562	1.625	1275	1295	1315	1336
400	.10	0.562	1.625	1177	1197	1217	1237
390	.12	0.670	1.625	1178	1198	1218	1238
390	.15	0.670	1.625	1179	1199	1219	1239
385	.18	0.670	1.875	1276	1296	1316	1336
385	.22	0.670	1.875	1180	1200	1220	1240
380	.27	0.750	2.375	1277	1297	1317	1337
380	.33	0.750	2.375	1183	1203	1223	1243
370	.39	1.000	1.875	1278	1298	1318	1338
370	.47	1.000	1.875	1181	1201	1221	1241
350	.56	1.000	2.375	1279	1299	1319	1339
350	.68	1.000	2.375	1182	1202	1222	1242
300	0.10	0.500	1.125	1044	1074	1104	1134
300	0.12	0.562	1.375	1340	1360	1380	1400
300	0.15	0.562	1.375	1045	1075	1105	1135
300	0.18	0.562	1.625	1341	1361	1381	1401
300	0.22	0.562	1.625	1046	1076	1106	1136
300	0.27	0.562	1.875	1342	1362	1382	1402
300	0.33	0.562	1.875	1047	1077	1107	1137
300	0.39	0.670	1.625	1343	1363	1383	1403
300	0.47	0.670	1.625	1048	1078	1108	1138
295	0.56	0.750	1.875	1344	1364	1384	1404
290	0.68	0.750	1.875	1049	1079	1109	1139
280	0.82	0.750	2.125	1345	1365	1385	1405
270	1.00	0.750	2.125	1050	1080	1110	1140

See footnote at end of table.

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TABLE I. Electrical characteristics, dimensions, and dash numbers - Continued.

AC rated voltage volts (rms)	Capacitance value µF	Dimensions, nominal (inches) 1/		Dash number Failure rate level for (%/1,000 hr)			
		D ±0.062	L +.031 -.051	M (1.0)	P (0.1)	R (0.01)	S (0.001)
235	1.50	1.000	1.875	1051	1081	1111	1141
200	2.20	1.000	2.625	1052	1082	1112	1142
165	0.10	0.312	0.875	1031	1061	1091	1121
165	0.12	0.312	1.125	1346	1366	1386	1406
165	0.15	0.312	1.125	1032	1062	1092	1122
165	0.18	0.400	0.875	1347	1367	1387	1407
165	0.22	0.400	0.875	1033	1063	1093	1123
165	0.27	0.400	1.125	1348	1368	1388	1408
165	0.33	0.400	1.125	1034	1064	1094	1124
165	0.39	0.400	1.375	1349	1369	1389	1409
165	0.47	0.400	1.375	1035	1065	1095	1125
165	0.56	0.562	1.125	1350	1370	1390	1410
165	0.68	0.562	1.125	1036	1066	1096	1126
165	0.82	0.562	1.375	1351	1371	1391	1411
165	1.00	0.562	1.375	1037	1067	1097	1127
155	1.50	0.562	1.625	1038	1068	1098	1128
150	2.20	0.670	1.625	1039	1069	1099	1129
145	2.50	0.670	1.875	1352	1372	1392	1412
140	3.30	0.750	1.875	1040	1070	1100	1130
135	4.00	0.750	2.125	1353	1373	1393	1413
130	4.70	0.750	2.375	1041	1071	1101	1131
110	6.80	1.000	1.875	1042	1072	1102	1132
100	8.00	1.000	2.125	1354	1374	1394	1414
90	9.00	1.000	2.375	1355	1375	1395	1415
80	10.00	1.000	2.625	1043	1073	1103	1133

1/ Metric equivalents see table II.

TABLE II. Metric equivalents of decimal inches. 1/

Inches	MM	Inches	MM	Inches	MM
.031	.79	.562	14.27	1.375	34.93
.051	1.30	.670	17.02	1.625	41.28
.062	1.57	.750	19.05	1.875	47.63
.312	7.92	.875	22.23	2.125	53.98
.400	10.16	1.000	25.40	2.375	60.33
.500	12.70	1.125	28.58	2.625	66.68

1/ Metric equivalents are given for general information only.

Temperature cycling: Method 102 of MIL-STD-202, condition D.

Seal:

For unimpregnated units: Method 112 of MIL-STD-202, condition A, except at 85°C +4°C, -0°C.

For impregnated units: One hour at rated temperature with no evidence of leakage.

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Dielectric withstanding voltage (DWV): Method 301 of MIL-STD-202.

Terminal to terminal: 140 percent of rated ac voltage (V rms at 60 Hz) on voltages of ≤300 volts; 120 percent of rated ac voltage (V rms at 60 Hz) on voltages of ≥350 volts.

Terminals to case: 900 V rms (at 60 Hz).

Insulation resistance (IR): Method 302 of MIL-STD-202, condition A.

Terminal to terminal: See table III.

Terminals to case: 10,000 megohms, minimum.

TABLE III. Insulation resistance.

In megohms:	
At +25°C ±3°C (need not exceed)	50,000
At +85°C +4°C, -0°C (need not exceed)	10,000
In megohms x microfarads (minimum):	
At +25°C ±3°C	25,000
At +85°C +4°C, -0°C	5,000

Capacitance (Cap.): Method 305 of MIL-STD-202.

Dissipation factor (DF):

At 25°C: 0.3 percent maximum.

At 85°C: 0.6 percent maximum.

Barometric pressure (reduced): Method 105 of MIL-STD-202.
120 percent of rated ac voltage (400 Hz) at .82 inch of mercury (80,000 feet).
See MIL-C-39022 for voltage limitations.

Vibration, high frequency: Method 204 of MIL-STD-202, condition B (15G).

Salt spray (corrosion): Method 101 of MIL-STD-202, condition B (48 hours).

Salt solution: 5 percent.

Immersion: Method 104 of MIL-STD-202, condition C.

DWV:

Terminal to terminal: 140 percent of rated voltage on voltages of ≤300 volts; 120 percent of rated voltage on voltages of ≥350 volts.

Terminals to case: 900 V rms (60 Hz).

IR:

Terminal to terminal: Not less than 33.3 percent of value specified in table III.

Terminals to case: 5,000 megohms, minimum.

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1. Audit schedules and frequency are at the discretion of the contractor

Cap.: Within ±10 percent of initial value.

DF: Not to exceed 110 percent of initial limit.

Solderability: Method 208 of MIL-STD-202.

Shock (specified pulse): Method 213 of MIL-STD-202, condition I. DC voltage equivalent to rms rating shall be applied.

Moisture resistance: Method 106 of MIL-STD-202.

DWV, IR, Cap., and DF: Same as for immersion.

Terminal strength: Method 211 of MIL-STD-202, conditions A and D.

Condition A: Applied force 5 pounds.

Condition D: 3 rotations of 360 degrees.

Low temperature and cap. change with temperature:

Low temperature: -55°C +0, -3°C for 48 ±4 hours with rated voltage applied.

Cap. change with temperature:

At -55°C +0, -3°C: -3 percent, maximum.

At 85°C +4, -0°C: -3 percent, maximum.

Fault count: Not applicable.

Fungus: Method 508 of MIL-STD-810, procedure I.

Resistance to soldering heat: Method 210 of MIL-STD-202, condition B (260 ±5°C for 10 ±1 seconds).

IR: See table III.

Cap.: Within ±5 percent of initial value.

DF: Not more than 1.0 percent.

Life: Method 108 of MIL-STD-202.

Qualification:

Accelerated conditions: 140 percent of rated ac voltage (400 Hz) for 2,000 +72, -0 hours on voltages of $\leq$300 volts; 120 percent of rated ac voltage (400 Hz) for 2,000 +72, -0 hours on voltages of $\geq$350 volts.

Rated conditions: 100 percent of rated voltage for 10,000 +96, -0 hours.

DF (at +85°C +4°C, -0°C) between 24 and 48 hours of test: Not greater than initial limit.

During last 48 hours of test: Not greater than 0.75 percent.

IR:

Terminal to terminal: Not less than 1,000 megohm/microfarads. Need not exceed 5,000 megohms.

Terminals to case: 5,000 megohms, minimum.

Cap.: Within ±10 percent of initial value.

DF (at +25°C ±3°C) after life: Not greater than 0.75 percent.

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TABLE IV. Substitutability data.

Dash no. in MIL-C-39022/7C				Substitutable for dash no. and type designation in:										
				MIL-C-39022/7B					MIL-C-39022/7A					MIL-C-39022/7
M	P	R	S	-	M	P	R	S	L	M	P	R	S	1/
1171	1191	1211	1231	1151	1171	1191	1211	1231	---	---	---	---	---	---
1270	1290	1310	1330	1250	1270	1290	1310	1330	---	---	---	---	---	---
1172	1192	1212	1232	1152	1172	1192	1212	1232	---	---	---	---	---	---
1271	1291	1311	1331	1251	1271	1291	1311	1331	---	---	---	---	---	---
1173	1193	1213	1233	1153	1173	1193	1213	1233	---	---	---	---	---	---
1272	1292	1312	1332	1252	1272	1292	1312	1332	---	---	---	---	---	---
1174	1194	1214	1234	1154	1174	1194	1214	1234	---	---	---	---	---	---
1273	1293	1313	1333	1253	1273	1293	1313	1333	---	---	---	---	---	---
1175	1195	1215	1235	1155	1175	1195	1215	1235	---	---	---	---	---	---
1274	1294	1314	1334	1254	1274	1294	1314	1334	---	---	---	---	---	---
1176	1196	1216	1236	1156	1176	1196	1216	1236	---	---	---	---	---	---
1275	1295	1315	1335	1255	1275	1295	1315	1335	---	---	---	---	---	---
1177	1197	1217	1237	1157	1177	1197	1217	1237	---	---	---	---	---	---
1178	1198	1218	1238	1158	1178	1198	1218	1238	---	---	---	---	---	---
1179	1199	1219	1239	1159	1179	1199	1219	1239	---	---	---	---	---	---
1276	1296	1316	1336	1256	1276	1296	1316	1336	---	---	---	---	---	---
1180	1200	1220	1240	1160	1180	1200	1220	1240	---	---	---	---	---	---
1277	1297	1317	1337	1257	1277	1297	1317	1337	---	---	---	---	---	---
1183	1203	1223	1243	1163	1183	1203	1223	1243	---	---	---	---	---	---
1278	1298	1318	1338	1258	1278	1298	1318	1338	---	---	---	---	---	---
1181	1201	1221	1241	1161	1181	1201	1221	1241	---	---	---	---	---	---
1279	1299	1319	1339	1259	1279	1299	1319	1339	---	---	---	---	---	---
1182	1202	1222	1242	1162	1182	1202	1222	1242	---	---	---	---	---	---
1044	1074	1104	1134	1014	1044	1074	1104	1134	1014	1044	1074	1104	1134	CHR49A1QD104K-
1304	1360	1380	1400	---	---	---	---	---	---	---	---	---	---	---
1045	1075	1105	1135	1015	1045	1075	1105	1135	1015	1045	1075	1105	1135	CHR49A1QD154K-
1341	1361	1381	1401	---	---	---	---	---	---	---	---	---	---	---
1046	1076	1106	1136	1016	1046	1076	1106	1136	1016	1046	1076	1106	1136	CHR49A1QD224K-
1342	1362	1382	1402	---	---	---	---	---	---	---	---	---	---	---
1047	1077	1107	1137	1017	1047	1077	1107	1137	1017	1047	1077	1107	1137	CHR49A1QD334K-
1343	1363	1383	1403	---	---	---	---	---	---	---	---	---	---	---
1048	1078	1108	1138	1018	1048	1078	1108	1138	1018	1048	1078	1108	1138	CHR49A1QD474K-
1344	1364	1384	1404	---	---	---	---	---	---	---	---	---	---	---
1049	1079	1109	1139	1019	1049	1079	1109	1139	1019	1049	1079	1109	1139	CHR49A1QZ684K-
1345	1365	1385	1405	---	---	---	---	---	---	---	---	---	---	---
1050	1080	1110	1140	1020	1050	1080	1110	1140	1020	1050	1080	1110	1140	CHR49A1QY105K-
1051	1081	1111	1141	1021	1051	1081	1111	1141	1021	1051	1081	1111	1141	CHR49A1QX155K-
1052	1082	1112	1142	1022	1052	1082	1112	1142	1022	1052	1082	1112	1142	CHR49A1QC225K-
1031	1061	1091	1121	1001	1031	1061	1091	1121	1001	1031	1061	1091	1121	CHR49A1QW104K-
1346	1366	1386	1406	---	---	---	---	---	---	---	---	---	---	---
1032	1062	1092	1122	1002	1032	1062	1092	1122	1002	1032	1062	1092	1122	CHR49A1QW154K-
1347	1367	1387	1407	---	---	---	---	---	---	---	---	---	---	---
1033	1063	1093	1123	1003	1033	1063	1093	1123	1003	1033	1063	1093	1123	CHR49A1QW224K-
1348	1368	1388	1408	---	---	---	---	---	---	---	---	---	---	---
1034	1064	1094	1124	1004	1034	1064	1094	1124	1004	1034	1064	1094	1124	CHR49A1QW334K-
1349	1369	1389	1409	---	---	---	---	---	---	---	---	---	---	---

See footnote at end of table.

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TABLE IV. Substitutability data - Continued.

Dash no. in MIL-C-39022/7C				Substitutable for dash no. and type designation in:										
				MIL-C-39022/7B					MIL-C-39022/7A					MIL-C-39022/7
M	P	R	S	L	M	P	R	S	L	M	P	R	S	1/
1035	1065	1095	1125	1005	1035	1065	1095	1125	1005	1035	1065	1095	1125	CHR49A1QW474K-
1350	1370	1390	1410	---	---	---	---	---	---	---	---	---	---	---
1036	1066	1096	1126	1006	1036	1066	1096	1126	1006	1036	1066	1096	1126	CHR49A1QW684K-
1351	1371	1391	1411	---	---	---	---	---	---	---	---	---	---	---
1037	1067	1097	1127	1007	1037	1067	1097	1127	1007	1037	1067	1097	1127	CHR49A1QW105K-
1038	1068	1098	1128	1008	1038	1068	1098	1128	1008	1038	1068	1098	1128	CHR49A1QU155K-
1039	1069	1099	1129	1009	1039	1069	1099	1129	1009	1039	1069	1099	1129	CHR49A1QV225K-
1352	1372	1392	1412	---	---	---	---	---	---	---	---	---	---	---
1040	1070	1100	1130	1010	1040	1070	1100	1130	1010	1040	1070	1100	1130	CHR49A1QT335K-
1353	1373	1393	1413	---	---	---	---	---	---	---	---	---	---	---
1041	1071	1101	1131	1011	1041	1071	1101	1131	1011	1041	1071	1101	1131	CHR49A1QS475K-
1042	1072	1102	1132	1012	1042	1072	1102	1132	1012	1042	1072	1102	1132	CHR49A1QR685K-
1354	1374	1394	1414	---	---	---	---	---	---	---	---	---	---	---
1355	1375	1395	1415	---	---	---	---	---	---	---	---	---	---	---
1043	1073	1103	1133	1013	1043	1073	1103	1133	1013	1043	1073	1103	1133	CHR49A1QP106K-

1/ Complete type designation will include an additional symbol for failure rate level.

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Quality conformance: 240 hours at accelerated conditions and meet the requirements for qualification test.

Extended life:

Accelerated-conditions: 1,760 +72, -0 hours.

Marking: In accordance with MIL-C-39022.

Part number: M39022/07- (dash number from table I).

Revision letters are not used to denote changes due to the extensiveness of the changes.

Custodians:
Army - ER
Navy - EC
Air Force - 11

Review activities:
Army - AR
Air Force - 17, 99
DLA - ES

User activities:
Navy - AS, MC, OS, SH, CG
Air Force - 19

Agent:
DLA - ES

Preparing activity:
Navy - EC

(Project 5910-1506)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-STD-70C	Abrasives and Plastic Blasting

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*:
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording

c. Reason/Rationale for Recommendat

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

DD FORM 82 MAR 1426 PREVIOUS EDITION IS OBSOLETE.

(TO DETACH THIS FORM, CUT ALONG THIS LINE.)

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